

MOJAVE

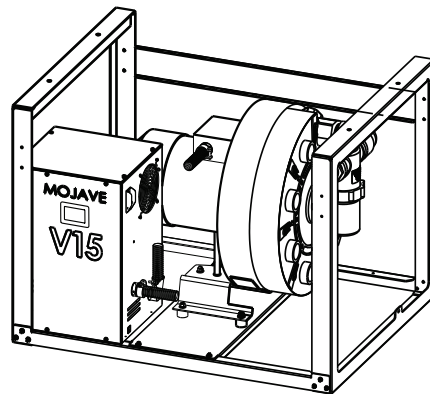
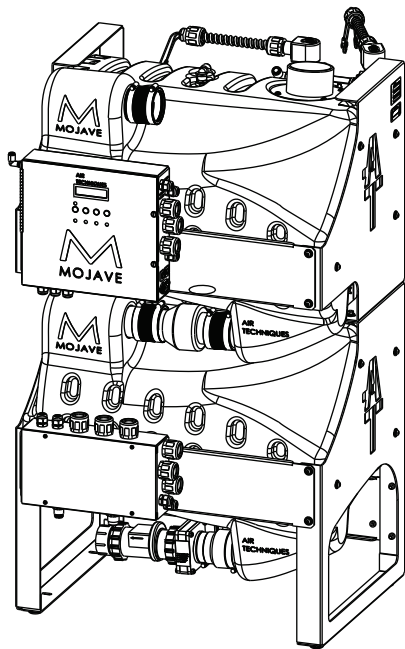
DRY VACUUM SYSTEM

Part Numbers V15, 2V15, 3V15 and 4V15

PRE-INSTALLATION GUIDE

All pumps comply with NFPA 99C level 3 requirements.

All installations must conform to local codes.



System being installed: (AS CHECKED)



V15



3V15



2V15



4V15

Doctor: _____

Address: _____

Phone#: _____

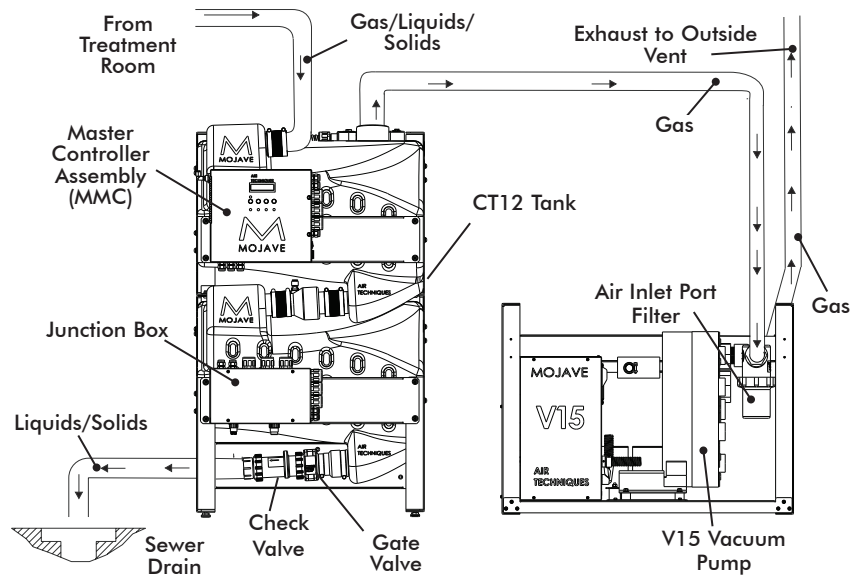
Dealer: _____

Dealer Address: _____

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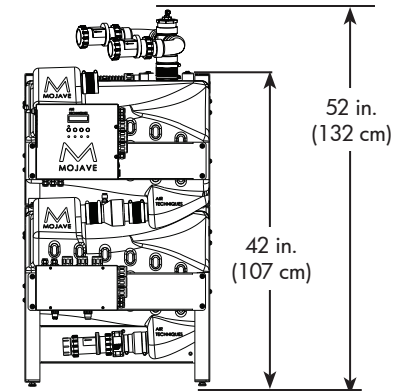
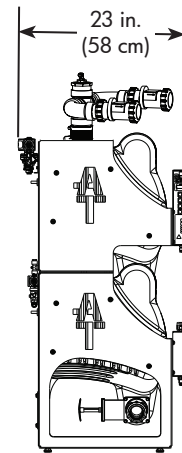
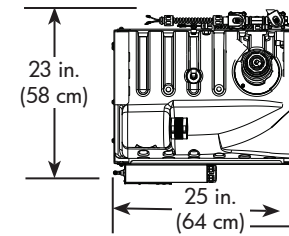
SYSTEM COMPONENTS & DIMENSIONS

System Components				
	V15	2V15	3V15	4V15
V15 Pump Assembly	1	2	3	4
CT12 Tank Assembly	1	1	1	1
CT34 Tank Assembly	0	0	1	1
Maximum Users	15	30	45	60

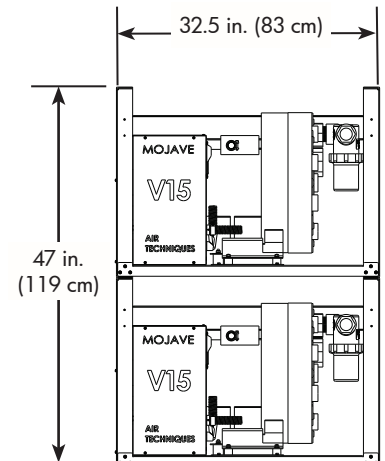
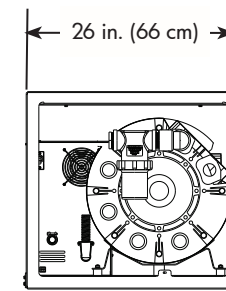
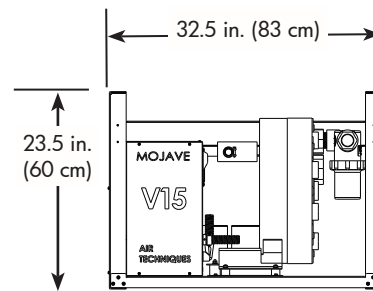


Typical V15 MOJAVE System Installation

Physical Characteristics			
	CT12 & CT34 Continuum Tank	One V15 Pump	Two V15 Pumps Stacked
Width	25 in. (64 cm)	32.5 in. (83 cm)	32.5 in. (83 cm)
Depth	23 in. (58 cm)	26 in. (66 cm)	26 in. (66 cm)
Height	42 in. (107 cm)	23.5 in. (60 cm)	47 in. (119 cm)
Weight	175 Lbs. (65 kg)	250 Lbs (93 kg)	500 Lbs (187 kg)



CT12 and CT34 Continuum Tank Dimensions



One Pump

Two Stacked Pumps

V15 Vacuum Pump Dimensions

MOJAVE SYSTEM CONFIGURATIONS

Important:

- ☐ Side by side installation of pump and tank is preferred.
- ☐ V15 pumps should only be stacked two high in all other system configurations as shown.
- ☐ All tanks are shipped with leveling feet set to lowest position.
- ☐ V15 pumps are shipped without leveling feet. They are supplied in pump accessory kit.

Note: Stacked V15 System is only configuration that allows a CT12 Tank to be stacked above a V15 Pump. Use Drip Shield Kit, H5454

Minimum System Footprint

Side by Side V15 System Installation

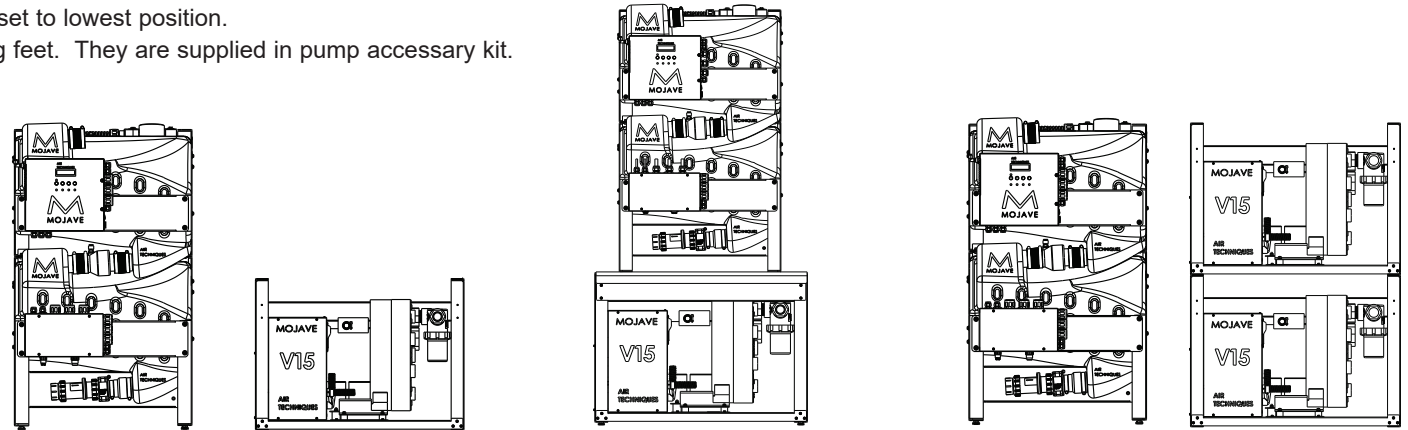
Width	Depth	Area
64 in. (162.5 cm)	30 in. (76.2 cm)	13.25 ft ² (1.2 m ²)

Stacked V15 System Installation

Width	Depth	Area
36 in. (91.5 cm)	30 in. (76.2 cm)	7.5 ft ² (0.7 m ²)

2V15 System Installation

Width	Depth	Area
64 in. (162.5 cm)	30 in. (76.2 cm)	13.25 ft ² (1.2 m ²)



V15 System Installation: One V15 Pump and One CT12 Tank Installed Side by Side

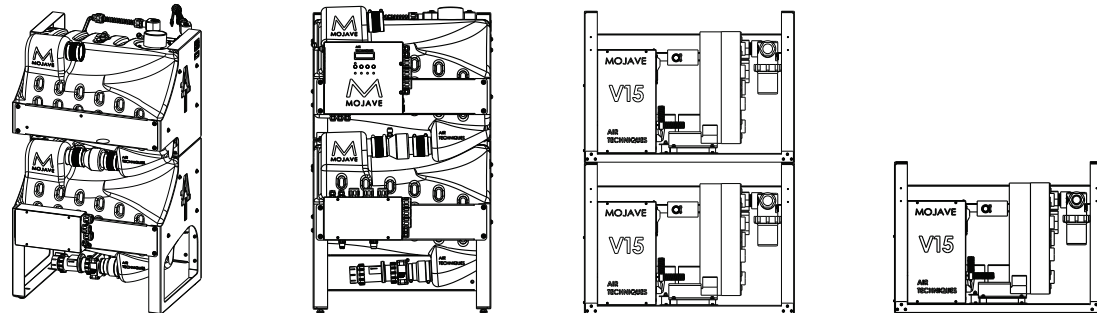
Stacked V15 System Installation: One CT12 Tank above One V15 Pump (See note.)

2V15 System Installation: Two V15 Pumps Stacked with One CT12 Tank on the Side

Minimum System Footprint

3V15 System Installation

Width	Depth	Area
130 in. (330.2 cm)	30 in. (76.2 cm)	27 ft ² (2.5 m ²)

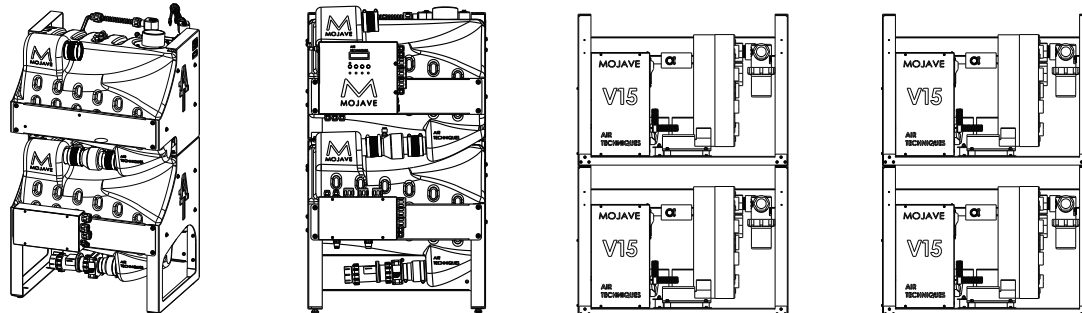


3V15 System Installation Three V15 Pumps (2 stacked) with CT12 and CT34 Tanks on the Side

Minimum System Footprint

4V15 System Installation

Width	Depth	Area
130 in. (330.2 cm)	30 in. (76.2 cm)	27 ft ² (2.5 m ²)

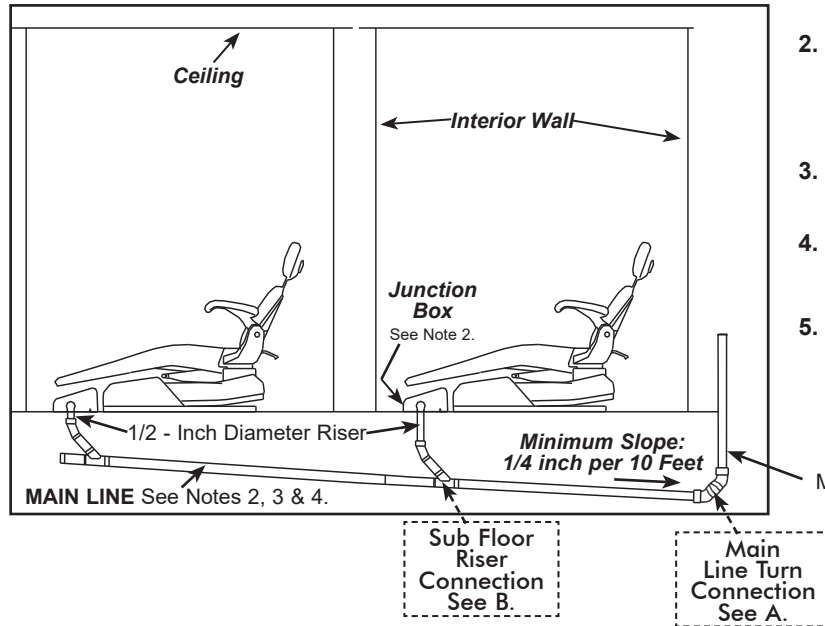


4V15 System Installation Two Side by Side Stacks of Two V15 Pumps with CT12 and CT34 Tanks on the Side

TREATMENT ROOM PLUMBING INSTALLATIONS

SUB FLOOR INSTALLATION -

Recommended system installation layout should be used whenever possible.



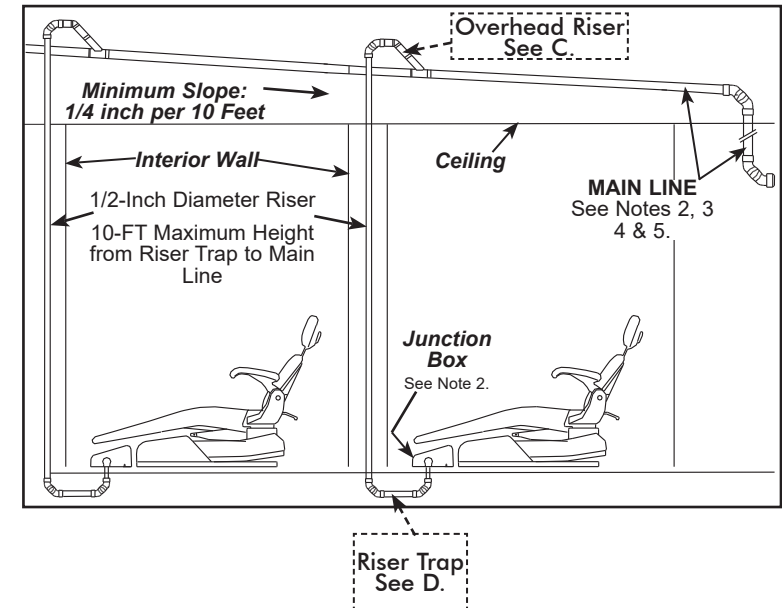
Notes:

1. 10-foot Maximum Height from Main Line to Tank.
2. Consult Dental Unit Manufacturer's Guidelines for correct reduced size and height of termination of vacuum line inside junction box.
3. Limit branches. Orient main line under junction box or cabinet.
4. When piping line is above 3/4" I.D. or larger, use 45° WYE's & elbows only.
5. Recommend installing separate line connection for scavenger when using Nitrous scavengers in overhead piping installations.

Main Line Riser for connection to tank input. See Note 1.

OVERHEAD INSTALLATION -

Alternate system installation layout should be used only when unable to use the sub-floor plumbing layout.

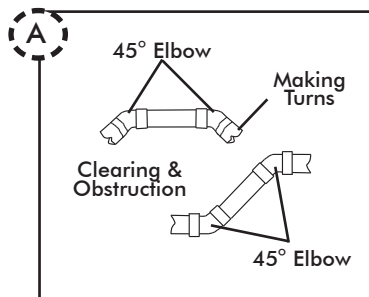


CONNECTOR DETAILS - ALL INSTALLATIONS

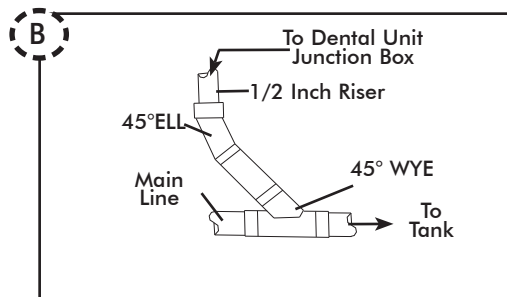
- ☐ Use only 45° elbows to make turns in main line.
- ☐ Make sure to use the proper pipe type for associated system.
- ☐ If piping is diverted to clear an obstruction, **DO NOT MAKE A TRAP.** See detail A, Main Line Turn Connections.
- ☐ DO NOT use standard 90° elbows.

Important:

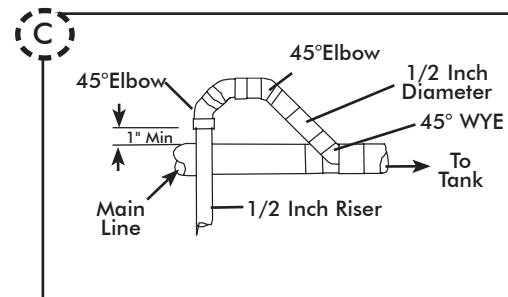
All installation pipes and fittings provided by plumber.
All installations must conform to local codes.



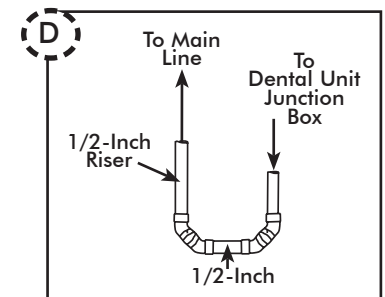
Main Line Turn Connections



Sub Floor Riser to Main Line Detail

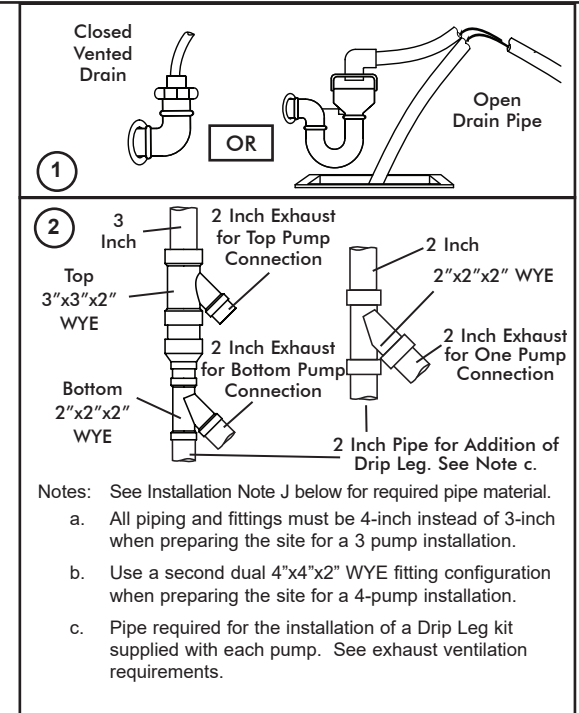
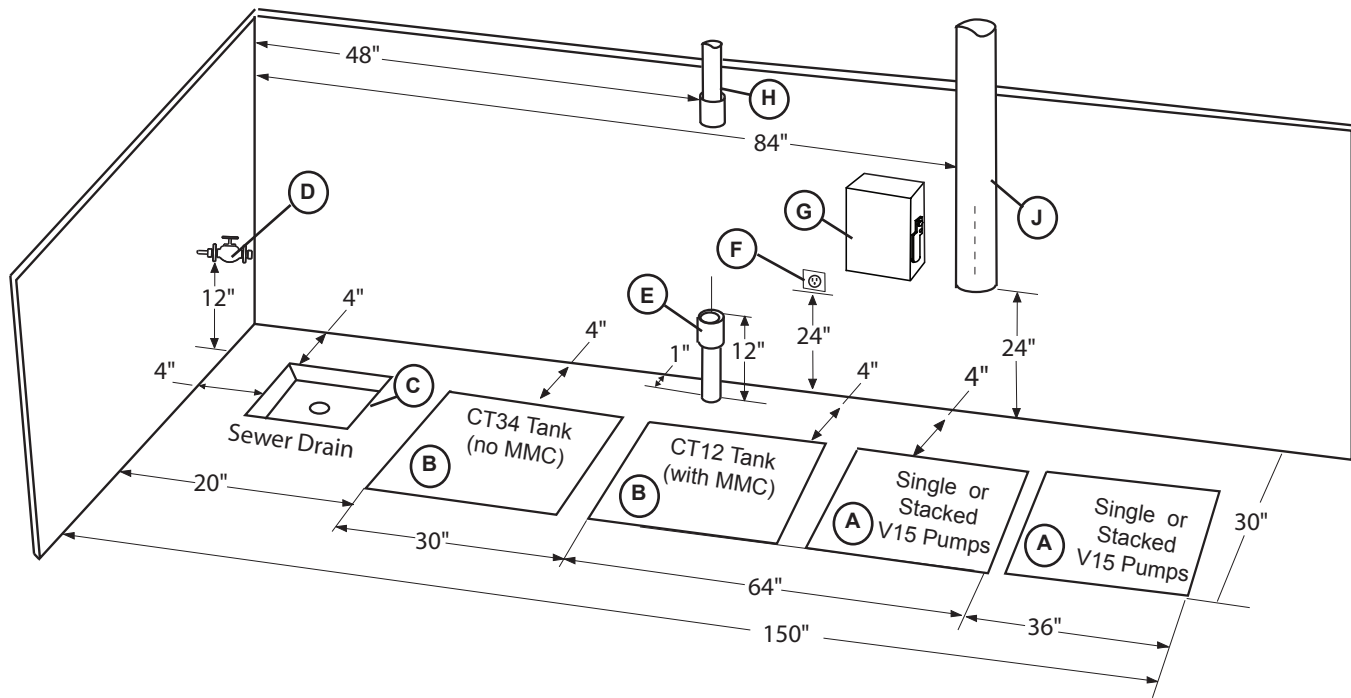


Overhead Riser to Main Line Detail
(Prevents liquids from draining down the 1/2" riser.)



Riser Trap Detail (45° Elbows)

TYPICAL EQUIPMENT ROOM FLOOR PLAN LAYOUT



INSTALLATION NOTES:

- PUMP INSTALLATION SPACE** - Area for single or stacked V15 pumps in typical side by side installations. Only stack up to 2 pumps in one area.
- TANK INSTALLATION SPACE** - Area for CT12 or CT34 tank in typical side by side installations.
- SEWER DRAIN** - Provide a drain for the removal of waste liquids from the MOJAVE tank. Use an open drain pipe (1 1/2 inch P-Trap with 1 inch air gap or floor sink) or a closed vented drain.
- TANK WASHOUT** - Provide a water source terminated with a 1/2 inch FNPT shut-off valve providing water pressure between 20 and 100 psi for daily tank washout. Valve location must be no more than 10 feet from the tank installation to allow connection of supplied 10-foot 3/8-inch Poly tubing to the tank washout port. Provisions for backflow prevention may be required. Check local code requirements.
- SUB FLOOR INSTALLATION VACUUM LINE** - See Plumbing Requirements for connection to tank inlet via supplied hose.
- MASTER CONTROLLER ELECTRIC OUTLET** - Master Controller requires a dedicated stand alone single phase 220V, hospital grade grounded receptacle. The supplied 10-foot line cord is the Mains disconnect device for the unit.
- PUMP ELECTRIC SERVICE** - Each MOJAVE pump is wired directly with a dedicated 220V, 40 AMP, three phase 50/60 Hz circuit. If Main Circuit panel is not located in equipment room, a disconnect box with approved ground is needed for each pump. Disconnect boxes should be mounted no more than 3 feet of each other and 3 feet of installation center line.
- OVERHEAD INSTALLATION VACUUM LINE** - See Plumbing Requirements for connection to tank inlet via supplied hose.
- HEAT EXHAUST** - Refer to Exhaust Line Options and Exhaust Ventilation Requirements for the exhaust vent line required for specific **MOJAVE** configurations. Schedule 40 metal pipe can normally be used on typical **MOJAVE** configuration installations. When installing two or more pumps, a reducing WYE adapter is needed to connect each vent tube to a common 3-inch exhaust vent line.

SITE REQUIREMENTS

Electrical	V15	2V15	3V15	4V15	Master Controller
Voltage Rating Volts AC	All pumps 220 Volts 3 Phase AC, 60 Hz, $\pm$ 10%				220 (Single Phase $\pm$ 10%)
Voltage Minimum/Maximum	198/242 Volts AC All pumps				205/240
Wire Size AWG Minimum Gauge	#8 AWG	#8 AWG	#8 AWG	#8 AWG	#14 AWG
Minimum Circuit Breaker Rating	40A	40A (Qty 2)	40A (Qty 3)	40A (Qty 4)	15A
Incoming Power	Hard wire Connection (Each pump is supplied a 6 foot BX cable)				NEMA 6-15R (Supplied 10-ft. line cord)
Remote (Low Voltage Wiring)	#18 AWG (Qty 4) Wire Connection between the MMC and the Remote Switch Panel . (See Figure 16, page 26.)				

Plumbing	V15	2V15	3V15	4V15
Exhaust Vent Pipe (See note 1)	2" Metal Pipe	One 3" or two 2" Metal Pipe	One 4" or three 2" Metal Pipe	Two 3" or four 2" Metal Pipe
Minimum Suction Line Pipe	2" PVC Sch. 40	3" PVC Sch. 40	3" PVC Sch. 40	4" PVC Sch. 40
Maximum Suction Line Pipe (See note 2)	3" PVC Sch. 40	4" PVC Sch. 40	4" PVC Sch. 40	6" PVC Sch. 40
Minimum Riser Pipe	½" PVC Sch. 40	½" PVC Sch. 40	½" PVC Sch. 40	½" PVC Sch. 40
Vacuum Line Termination	2"	2"	2"	2"
Branch Line Pipe	Size requirement of Branch piping differs by the number of operatories being serviced. Up to two operatories use 1" PVC Schedule 40. Three to six operatories use 1 ½" PVC Schedule 40. More that six operatories use 2" PVC Schedule 40			
Drain Line Pipe	1 ½" PVC Schedule 40			
Wash-Out Water Line	½" FNPT Shut-off Valve			

NOTES

1. Recommended for all new installations.
2. Use maximum internal diameter for the main line when preparing any new installation.

EXHAUST VENTILATION REQUIREMENTS

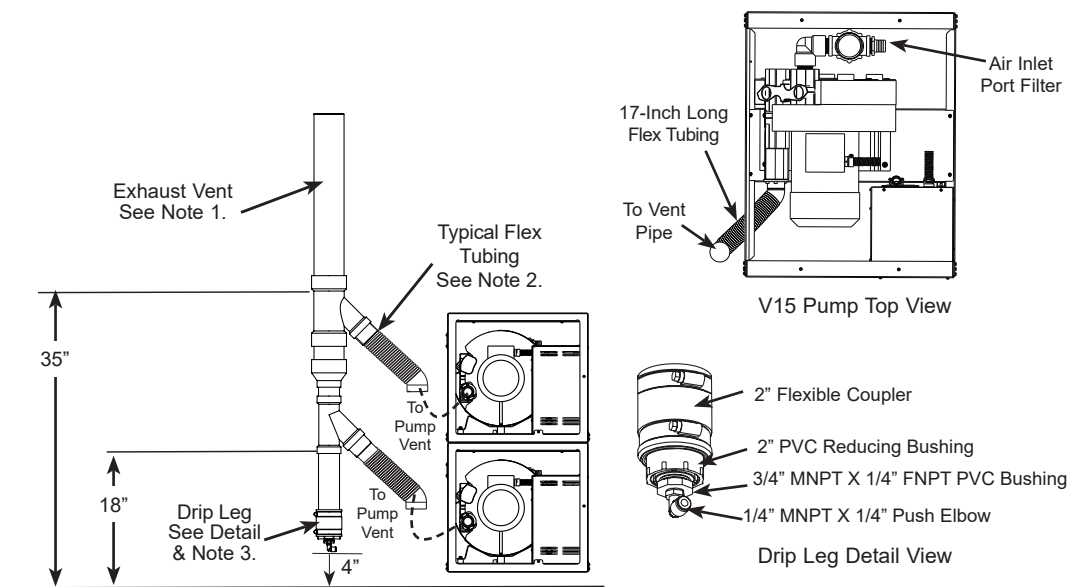
HEAT EXHAUST CONNECTION NOTES

- 1. VENT LINE** - Refer to Exhaust Line Options and use metal pipe (supplied by installer) to fabricate the required exhaust vent line for MOJAVE systems. Do not make a trap in the exhaust vent piping.

Also see Exhaust Vent Protection and Ventilation Requirements below.

- 2. V15 PUMP EXHAUST VENT CONNECTION** - Connection between the pump and exhaust vent piping is typically made via the supplied 2-inch Black Flex tubing.

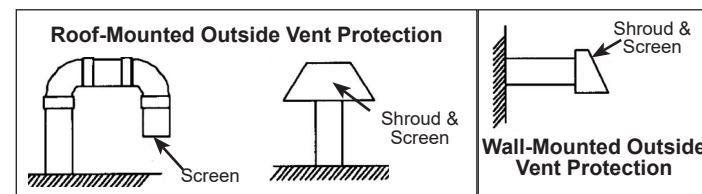
- 3. DRIP LEG** - The supplied drip leg must be installed at the lower end of the vent pipe to collect condensation produced during pump operation. The bottom of drip leg should be located a minimum of 4 inches from floor. Attach the drain tube to the drip leg quick-connect fitting to allow drainage into floor drain/sink.



V15 Pump Heat Exhaust Connection

Exhaust Vent Protection.

If the exhaust piping is venting to the outside of the building, precautions must be taken to protect the equipment room from weather elements and animal intrusion. This can be accomplished by using one of the three methods shown on the right.



Exhaust Vent Requirements.

The MOJAVE equipment must be used in a controlled-temperature environment. Maintain equipment room temperature between 40 and 105 degrees Fahrenheit. An exhaust fan is necessary if room temperature is not maintained by other methods.

Adequate forced ventilation must be provided across the unit by placing an appropriate exhaust fan opposite an equivalent air intake vent. The fan should be placed higher than the associated intake vent. Recommended minimum exhaust fan requirements for each MOJAVE unit are listed to the right.

Mojave Unit	Min (Watts)	Max (Watts)	Min (BTU/hr)	Max (BTU/hr)
V15	1,219	5,892	4,159	20,097
2V15	2,438	11,784	8,319	40,195
3V15	3,657	17,676	12,478	60,292
4V15	4,876	23,568	16,638	80,390

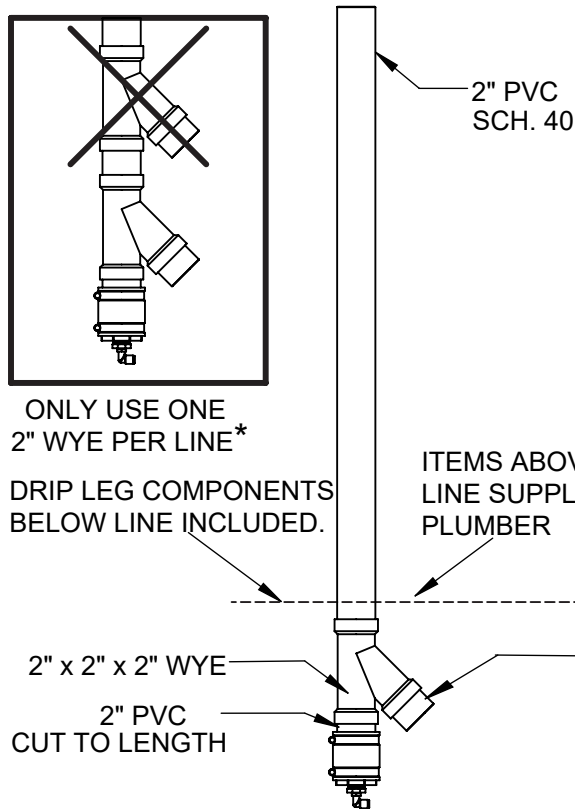


MEDICAL ELECTRICAL EQUIPMENT

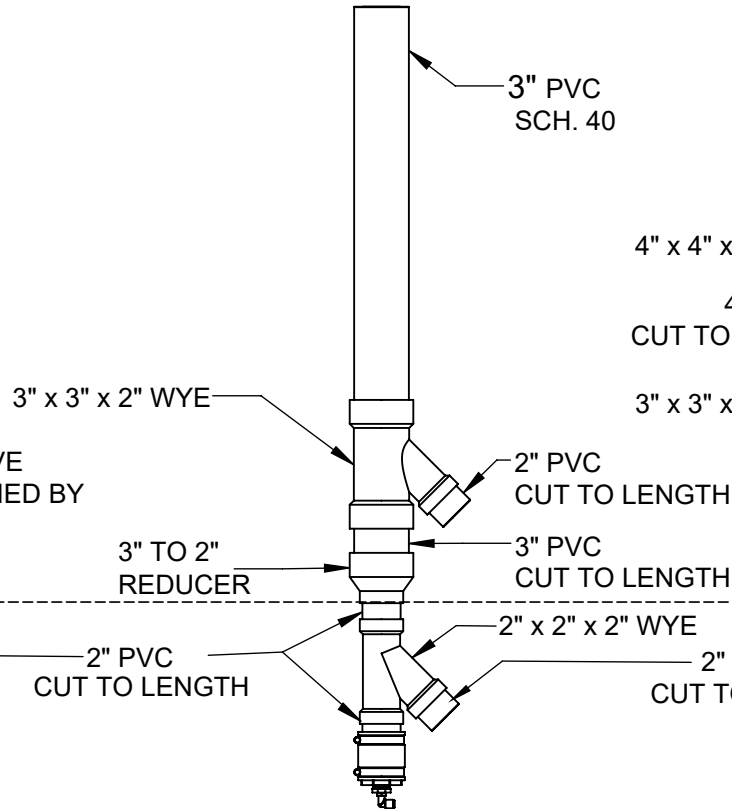
WITH RESPECT TO ELECTRICAL SHOCK, FIRE, MECHANICAL
AND OTHER SPECIFIED HAZARDS ONLY
IN ACCORDANCE WITH UL-60601-1, CAN/CSA C22.2 NO.601.1 66CA

MOJAVE EXHAUST LINE OPTIONS

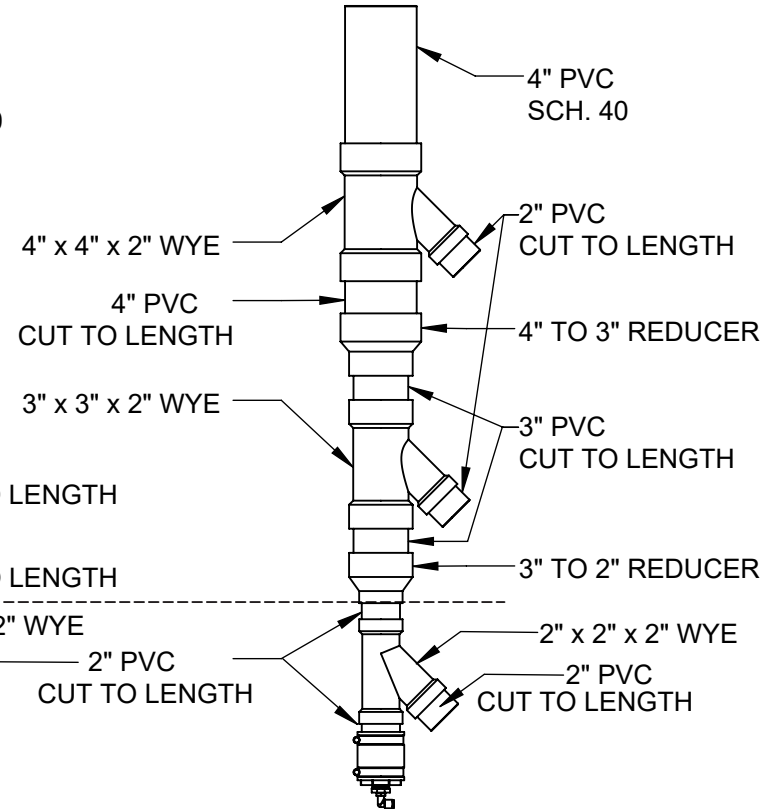
2" EXHAUST LINE



3" EXHAUST LINE



4" EXHAUST LINE



DRIP LEG ASSEMBLY PROVIDED (1 PER PUMP)

2" EXHAUST LINE	
MODEL#	QTY
V3, V5, V5	1x
2V3, 2V5, 2V7	2x
3V5	3x
4V5	4x

3" EXHAUST LINE	
MODEL#	QTY
2V3, 2V5, 2V7	1x
4V5	2x

4" EXHAUST LINE	
MODEL#	QTY
3V5	1x

***NOTE:** Do not use multiple 2" WYE assemblies together with a single 2" exhaust line for multi-pump systems. Make sure to take into account pump locations before assembling manifold.

NOTES

NOTES

NOTES

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- Waterline Cleaner
- Evacuation System Cleaner
- Imaging Accessories
- Chemistry
- Processor Accessories

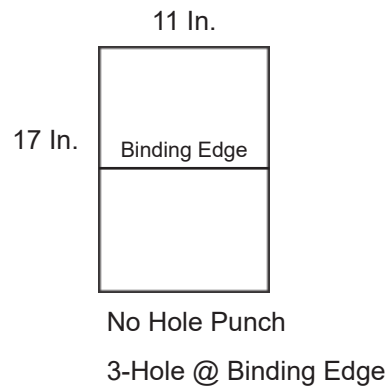
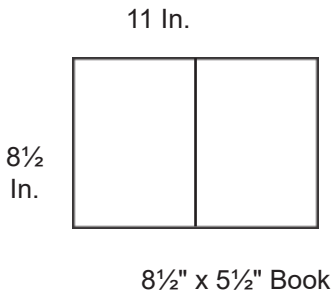
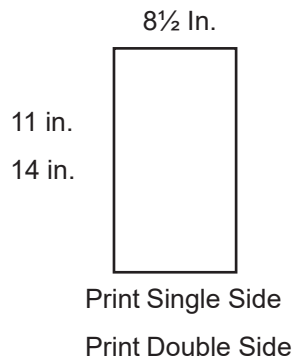
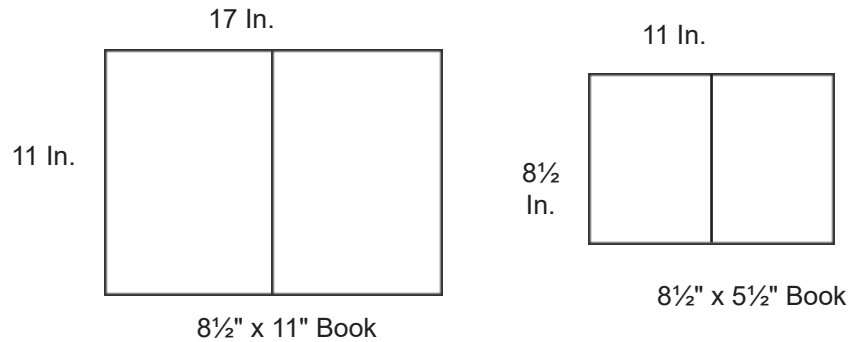
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Additional Options:

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Card Stock
Saddle Stitch / Fold (Staple Binding).
UV Coat Both Sides
Laminate
3-Hole Binding
Single hole punch left corner front side
Spiral Bound Booklets
Wire-O Booklets (Metal Coil Binding)

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